DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
BACHELORS OF SCIENCE IN COMPUTER SCIENCE

Course Title and Course Code: Knowledge Based System (CT-359)

Course Learning Outcome:

CLO	Mapping to the Project	How the Project Addresses the CLO
CLO 1(C2, PLO-2)	Knowledge Representation	Students understand and construct representations of equipment, components, symptoms, faults, causes, evidence, and maintenance actions using an appropriate formalism such as rules, semantic networks, frames, ontologies, or knowledge graphs.
CLO 2 (C4, PLO-3)	Problem Analysis & Solution Design	Students analyze the industrial fault-diagnosis problem, identify requirements and stakeholders, determine relationships among different knowledge components, and design a KBS solution using an appropriate knowledge-representation formalism.
CLO 3 (C5, PLO-4)	Method and Algorithm Assessment	Students assess alternative knowledge-representation and inference methods, such as forward chaining, backward chaining, rule-based reasoning, ontology-based reasoning, or graph-based reasoning, and justify the most appropriate approach for the given problem.

Complex Problem solving Attributes (CPA): (NCEAC Manual 2021 https://www.seoulaccord.org/document.php?id=79)

- **CPA-2:** Depth of Analysis Required: The problem does not have a single obvious solution and requires students to apply conceptual thinking, critical analysis, and innovative approaches to develop an appropriate abstract model and solution.
- **CPA-3:** Depth of Knowledge Required: Solving the problem requires in-depth computing and domain-specific knowledge, together with analytical reasoning and the application of well-established principles and techniques.
- **CPA-9:** Requirement Identification: The problem contains incomplete or unclear requirements, requiring students to identify, analyze, and define the relevant requirements and determine the underlying cause of the problem.

Problem Statement:

- **Explain** how a Knowledge-Based System can represent industrial equipment knowledge, including equipment, components, symptoms, faults, causes, and maintenance actions, using an appropriate knowledge representation formalism. (CLO 1)
- **Analyze** the given industrial equipment fault-diagnosis problem by identifying incomplete requirements, stakeholder needs, relationships, constraints, and suitable knowledge-based approaches for developing an effective solution. (CLO 2)
- **Evaluate** suitable knowledge representation methods and inference algorithms for diagnosing equipment faults and recommending maintenance actions, and justify the selected approach based on accuracy, explainability, safety, cost, and system requirements. (CLO 3)

Problem Description:

A manufacturing organization operates different types of industrial equipment and faces difficulties in diagnosing faults and selecting appropriate maintenance actions. Equipment-related knowledge is distributed across maintenance manuals, operator observations, historical maintenance records, expert experience, and operational or sensor information. The organization requires a **Knowledge-Based System (KBS)** that can represent this heterogeneous knowledge and assist maintenance personnel in identifying possible equipment faults based on observed symptoms, available evidence, equipment conditions, and previously known fault–cause relationships. Recent KBS research similarly highlights the use of knowledge graphs, ontologies, and knowledge-driven methods to organize complex equipment operation and maintenance knowledge and support intelligent decision-making.

The proposed system must handle **incomplete and potentially conflicting information** and support different stakeholder requirements, including safety, maintenance cost, equipment availability, and production continuity. Students are required to analyze the problem, identify the relevant knowledge and stakeholders, select and justify an appropriate **knowledge representation formalism and inference mechanism**, and develop a prototype that can infer possible faults and recommend suitable maintenance actions. The system should also provide an explanation of how its conclusions were reached, while considering constraints such as safety, downtime, and maintenance priorities.

Instructions:

- Students may work in group of 5 at maximum.
- Your assignment will be graded according to the rubrics (provided with the assignment).
- Submit your assignment on Google Classroom.

Deliverables:

1. **Dataset + Source Code**
2. Submit the report + presentation according to the **prescribed template before deadline**.
3. **Project Demonstration and Viva:** Students must demonstrate the implemented system.
4. Students must be able to explain and defend their **problem analysis, knowledge representation, reasoning/inference mechanism, implementation decisions, dataset/knowledge base, and results**.

Deadline for online project submission on google classroom is 18 October 2026. Presentation will be in week 10. The slot will be announced on GCR.

Complex Computing Problem Assessment Rubrics

Knowledge Based System (CT-359)

Name: _____ **Roll No.:** _____

Project Title: _____

Criteria	Excellent (2)	Average (1)	Poor (0)
Criterion 1: Understanding the Problem (CLO 2 / CPA-9)	Clearly understands the problem and accurately identifies requirements, stakeholders, constraints, and underlying issues.	Partially understands the problem and identifies some requirements or issues.	Fails to define the problem adequately and does not identify the underlying requirements or issues.
Criterion 2: Research and Knowledge Acquisition (CLO 1 / CLO 3 / CPA-3)	Uses relevant research extensively and develops a comprehensive knowledge base from appropriate sources.	Uses limited research with incomplete or partially relevant knowledge.	No apparent research or inadequate knowledge acquisition.
Criterion 3: Proposed KBS Approach and Knowledge Representation (CLO 1 / CLO 2 / CPA-2)	Proposes an appropriate and well-justified KBS approach with effective knowledge representation and reasoning.	Approach is partially appropriate but has weak justification or representation.	Approach is inappropriate, unclear, or lacks a suitable knowledge representation.
Criterion 4: Implementation and Code (CLO 3 / CPA-3)	Complete, correct, and functional implementation that effectively realizes the proposed solution.	Partially implemented solution with significant errors or missing functionality.	Incomplete, incorrect, or non-functional code.
Criterion 5: Report, Presentation and Documentation (CLO 2 / CLO 3 / CPA-2)	Complete, well-organized report and presentation, following the prescribed template and submitted on time.	Incomplete or poorly organized documentation with partial compliance with the template.	Report/presentation is missing, substantially incomplete, or not according to the required template.

Total marks: _____**/10**_____

Teacher's signature: _____